



**Service and Operational Optimization of AMRIT Pharmacy at
KGMU**

**BUSINESS DATA MANAGEMENT-CAPSTONE
PROJECT-MIDTERM REPORT**

SUBMITTED BY:

Neeraj Kumar

23F2002096@ds.study.iitm.ac.in

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IITM BS Online Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

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Declaration Statement

I am working on a Project titled “**Service and Operational Optimization of AMRIT Pharmacy at KGMU**”. I extend my appreciation to **Amrit Pharmacy**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Neeraj Kumar

Signature of Candidate

Name: Neeraj Kumar

Date: 08/09/2025

1. **Executive Summary:**

Amrit Pharmacy, located inside King George's Medical University (KGMU), Lucknow, has long been known for providing affordable medicines and medical supplies to patients and visitors. By purchasing directly from manufacturers and operating within the hospital premises, the pharmacy minimizes costs and ensures accessibility. However, with a growing customer base and thousands of daily transactions, challenges such as stock-outs, fluctuating sales patterns, and inefficient billing practices have started to affect smooth operations.

To address these issues, sales data spanning 56,281 rows and 21 attributes was digitized, cleaned, and consolidated into a structured format. This transformation allowed the identification of clear sales trends across items, categories, and time periods. Descriptive statistics revealed stable daily demand for essential drugs such as Amlorin and Diamox, contrasted with high variability in products like 3M Avagard Handrub, where bulk returns created unusual spikes and dips.

Further analysis highlighted category-level movements: gastroenterology items showed rising demand, orthopaedics remained stable, while cardiology products declined slightly. Weekday patterns revealed Mondays and Tuesdays as peak sales days, with Sundays recording the lowest activity. ABC segmentation grouped products into high-, medium-, and low-priority categories, providing a practical framework for optimizing stock levels and reducing expiry-related waste.

Overall, the findings show that while Amrit Pharmacy has succeeded in keeping essential healthcare affordable, operational improvements are necessary to ensure efficiency and continuity. The report builds on these insights, presenting systematic methods and evidence-based recommendations for strengthening inventory planning, streamlining billing, and supporting uninterrupted service delivery to the community.

2. **Proof of Originality:**

2.1 Details:

Outlet Name: Amrit Pharmacy

Type of Organisation: B2C (Business-to-Consumer)

Outlet Address: Shatabdi Phase I of King George's Medical University (KGMU), Lucknow

Outlet Employee Name: Arvindra Kushwaha

Employee Phone Number: 8860112432

Working Time: 00:00 to 23:59 (24 X 7)

2.2 Proof Links :

Interview Video:  Identification_video.mp4

Video of Shop:  Shop_video_proof.mp4

Image of Shop 1:  Shop_pic_1.jpg

Image of Shop 2:  Shop_pic_2.jpg

Image of Shop 3:  Shop_pic_3.jpg

Authorization Letter:  Authorization_Letter.pdf

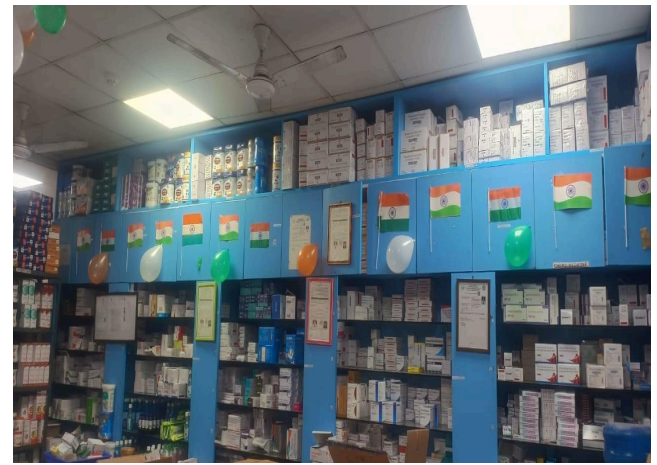


Image 1 & 2: Amrit Pharmacy KGMU, Lucknow

3. Meta Data:

The sales data was initially received in raw format, by system-generated, which required digitization. Through this process, a total of 8,563 individual bills spanning a 13-month period were compiled. To streamline the analysis, the digitized data was further consolidated, with daily sales aggregated into a single row for each day. This consolidation provided a clear summary of total sales for each SKU, making the dataset more manageable and suitable for further analysis.

Clean Dataset Link: [Dataset](#)

Data collection duration: 3 months

Data collection dates: 01/03/2025 to 31/05/2025

Dataset Dimensions: The dataset contains 21 columns and 56281 rows.

item name	item category	sale pk qty	mrp	selling rate (without tax)	e.p.r	e.p.r value	item gst %	sale value	discount
DRUNEM 1GM INJ [IW9764]	GENERAL MEDICINE	3	1067	366.678	203.7	611.13	12	1232.03	1969
DEQ10-FORTE [IK6156]	M2	1	450	318.6	270	270	18	375.94	74.06
TAXIM O 200MG TAB [I65111]	MEDICINES-DPCO/NPPA	1	109.49	88.35	78.19	78.19	12	98.95	10.54
DUPHALAC 150ML LEMONFLVR CANLAC [I87391]	MEDICINES-DPCO/NPPA	1	193.72	147.6	118.1	118.08	5	154.98	38.74
REFRESH TEARS EYE DROPS 10ML [I57331]	MEDICINES-DPCO/NPPA	1	137.64	110.141	97.47	97.47	12	123.34	14.3
DUPHALAC 150ML LEMONFLVR CANLAC [I87391]	MEDICINES-DPCO/NPPA	1	193.72	147.6	118.1	118.08	5	154.98	38.74
MEGAHEAL GEL 50GM [I40917]	M2	1	375	302.682	267.9	267.86	12	339	36
DEQ10-FORTE [IK6156]	M2	1	450	318.6	270	270	18	375.94	74.06
FLUMOL 650MG TAB [IT0994]	M2	1	22	11.24	6.61	6.61	12	12.58	9.42
FLUMOL 500 (PARACETAMOL) [IA9738]	M2	1	9.96	6.5	5	5	12	7.28	2.68

Image 3: How the data looks like (10 entries)

3.1 Columns: A sample of 10 columns is described below:

S.NO	COLUMN	DESCRIPTION	FORMAT
1.	ITEM NAME	This column contains the name of the item.	Text
2.	ITEM CATEGORY	Contains information about the category of the particular item (e.g., medicine, consumable, equipment).	Text
3.	SALE PK QTY	Represents the number of sale packs (larger units like strips, bottles, boxes) sold for the item.	Integer
4.	MRP	The maximum price printed on the item, which is legally allowed to be charged to a customer.	Float
5.	SELLING RATE (WITHOUT TAX)	The actual selling price of the item before adding GST or any other taxes.	Float
6.	E.P.R	The expected or calculated purchase rate at which the pharmacy bought the item from the supplier.	Float
7.	E.P.R VALUE	The total value derived from multiplying the E.P.R by the total quantity sold ($E.P.R \times \text{Quantity Sold}$).	Float
8.	ITEM GST %	The percentage of Goods and Services Tax (GST) applicable on the item.	Integer
9.	SALE VALUE	The total monetary value of the item sold, typically including taxes (MRP or $\text{Selling Rate} \times \text{Quantity Sold}$).	Float
10.	E.P.R VALUE	The total value derived from multiplying the E.P.R by the total quantity sold ($E.P.R \times \text{Quantity Sold}$).	Float

4. Descriptive Statistics:

	Mean	Standard Error	Standard Deviation	Mode	Median	Sample Variance	Kurtosis	Skewness	Count	Sum
quantity per box	5.97402	0.035863122	8.508029593	1	1	72.38656755	61.21444	6.1874684	56281	336224
sale pk qty	1.45637	0.009199835	2.182533552	1	1	4.763452706	4081.161	36.580001	56281	81966
sale ls qty	-0.87502	0.032500265	7.710238454	0	0	59.44777702	705.0104	-20.972376	56281	-49247
mrp	350.943	3.887387743	922.2289766	120	130	850506.2852	140.9951	10.713109	56281	19751416.1
selling rate (without tax)	175.35	2.136239015	506.7931604	75.63	80.766	256839.3074	118.9085	8.5020186	56281	9868883.59
e.p.r	129.918	1.488363251	353.0935961	70.95	61.6	124675.0876	91.2044	7.4167582	56281	7311935.83
e.p.r value	154.233	1.414089752	335.4732358	70.95	70.95	112542.292	95.11476	4.3505591	56281	8680360.37
basic selling price	205.101	1.834962443	435.3194606	75.63	91.35	189503.0328	87.82102	3.6356562	56281	11543280.4
item gst %	12.1374	0.011690257	2.773351739	12	12	7.69147987	2.724237	-0.2774579	56281	683105
tax amt	24.5918	0.225036529	53.38680398	9.06	10.72	2850.150839	80.91787	3.6073604	56281	1384049.46
sale value	229.693	2.04437364	484.9993709	84.69	103.94	235224.3898	85.26277	3.5344513	56281	12927349.5
mrp value	421.883	3.331001918	790.2341347	120	175	624469.9876	113.7635	7.7367979	56281	23743977.5
discount	192.19	2.420795098	574.3001555	25.31	44.7	329820.6686	635.7356	15.934527	56281	10816628
discount %	36.6613	0.152646299	36.21322328	23	23.01701	1311.397541	7.929658	2.806202	56281	2063337.11
total quantity sold	6.65383	0.073994803	17.55424353	1	3	308.1514659	157.0452	-4.6033534	56281	374484

Image 4: Descriptive Statistics Summary of the numeric Data

	Count	Unique Count	Mode	Mode Frequency
item name	56281	1379	DEPAN-40 MG TAB [IL5962]	657
pack	56281	36	1X1	26107
item category	56281	42	GENERAL MEDICINE	19141
item group	56281	5	MEDICINES	47261
payment type	56281	3	Cash	56150
weekday	56281	7	Tuesday	9250
analysis_category	1924	5	GASTROENTEROLOGY	993

Image 5: Descriptive Statistics Summary of the Categorical Data

4.1 Category Wise top 2 moving items:

4.1.1 CARDIOLOGY:

4.1.1.1 DIAMOX 250MG TABLETS [IF8802]:The average daily sales for Diamox 250Mg Tablets are 14.52 units, with a range from -90 to 60 units, indicating occasional large returns. A standard deviation of 19.33 shows high variability, mainly due to rare return events. The negative skewness of -3.12 supports this, reflecting a few days with significant negative movement. Most days had stable sales of 15 units, as shown by the 25th and 75th percentiles both at 15, suggesting consistent daily demand.

4.1.1.2 AMLOKIND 5MG TAB 15S [IK6954]:The average daily sales for Amlokind 5 Mg Tab 15S are 16.58 units, with a range from -45 to 27 units, indicating mostly stable sales with a few return instances. A standard deviation of 4.46 shows low variability, suggesting predictable daily demand. The negative skewness of -3.03 points to rare but significant returns affecting the overall distribution. The 25th percentile is 15 and the 75th percentile is 18, showing that most days experience consistent and moderate sales volumes.

4.1.2 GASTROENTEROLOGY:

4.1.2.1 DEPAN-40 MG TAB [IL5962]: The average daily sales for Depan-40 Mg Tab are 9.43 units, with a range from -20 to 60 units, indicating consistent demand with a few sharp return or surge events. A standard deviation of 4.76 shows moderate variability, with most values clustered around the mean. The slightly negative skewness of -0.62 suggests a few days with notable returns, but overall sales remain stable. The 25th percentile is 7 and the 75th percentile is 12, showing that most daily sales fall within this predictable range.

4.1.2.2 STENBAR DSR CAP [IK5792]: The average daily sales for Stenbar Dsr Cap are 11.12 units, with a range from -60 to 60 units, showing regular demand and occasional high returns or spikes. The standard deviation of 6.94 suggests moderate fluctuation in daily movement. A slight negative skewness of -1.63 points to occasional return events, but the overall trend is steady. The 25th percentile is 9 and the 75th percentile is 14, indicating that most days have moderate and stable sales volumes.

4.1.3 ORTHOPAEDICS:

4.1.3.1 ACEIX-SP TAB [IY4866]: The average daily sales for Aceix-Sp Tab are 11.33 units, with values ranging from -60 to 27, indicating consistent demand punctuated by occasional returns. A standard deviation of 6.15 reflects moderate daily variation, though most values stay near the average. The negative skewness of -2.87 reveals the presence of a few return-heavy days that influence the overall pattern. With the 25th percentile at 9 and the 75th at 14, most days show steady and predictable sales volumes.

4.1.3.2 DAXTON 8MG TAB [IY4867]: The average daily sales for Daxton 8Mg Tab are 12.81 units, with a wide range from -15 to 100, indicating mostly steady sales alongside occasional spikes and returns. The standard deviation of 7.04 shows moderate day-to-day variation, with most values clustering near the mean. A positive skewness of 4.17 highlights the presence of a few exceptionally high sales days. With the 25th percentile at 10 and the 75th percentile at 15, the majority of daily movements remain within a stable band.

5. Detailed Explanation of Analysis Process/Method:

5.1 Data Cleaning and Consolidation:

5.1.1 Process: The sales and return data for Amrit Pharmacy, located in Shatabdi Phase 1, KGMU, was received in Excel format, covering various medicines and medical items. During the initial review, issues such as missing some fields, duplicate rows, and some

unwanted columns for the analysis. These were resolved through data cleaning processes to ensure consistency and accuracy. After cleaning, the data was structured to aggregate daily sales for each item, merging multiple entries of the same product on the same day into a single summary row. This prepared dataset enabled effective analysis of product-wise trends and return patterns.

5.1.2 Justification: The Excel data from Amrit Pharmacy was thoroughly cleaned to address issues like missing values, duplicate records, and inconsistent formatting. Once refined, the dataset was reorganized to clearly present daily sales and return figures for each item. This transformation of raw data into a structured format made it easier to recognize sales trends, spot fast-moving or frequently returned products, and track demand fluctuations. With this organized information, the pharmacy could make more informed decisions regarding inventory management, procurement, and billing, ultimately leading to better operational efficiency and service delivery.

5.2 Descriptive Statistics Analysis:

5.2.1 Process: I analyzed the performance and movement trends of each product using a range of statistical measures, including the mean, range, standard deviation, skewness, and percentiles. These metrics gave me a comprehensive view of how each medicine performed over time in terms of sales and returns. For instance, I found that Diamox 250Mg Tablets had a consistent daily demand, with a mean of 14.52 units and low variability across percentiles. In contrast, 3M Avagard Handrub showed significant fluctuations, mainly due to bulk returns. This was clear from its wide value range, which spanned from -180 to 60 units, and its high standard deviation of 84.30. These statistical insights helped me identify which medicines had stable demand and which required more flexible inventory strategies.

5.2.2 Justification: This stage of my analysis provided valuable insights into the individual behavior of each medicine. For essential drugs such as Amlokind, Dytor Plus, and Diamox, I observed a consistent daily demand pattern, which supports maintaining stable stock levels and reduces the risk of overstocking. On the other hand, items like 3M Avagard Handrub and Daxton showed significant fluctuations, often caused by bulk purchases or returns, indicating the need for more flexible inventory management strategies. By applying descriptive statistics, I was able to help Amrit Pharmacy identify high-movement products, optimize shelf space, and improve procurement planning to boost overall operational efficiency.

5.3 Trend Analysis:

5.3.1 Process: An in-depth analysis was conducted on the daily, weekly, and monthly sales patterns at Amrit Pharmacy to gain a clearer understanding of how customer demand fluctuates over different time periods. Daily sales varied quite a bit- some days saw more than 7,000 items sold, while others dipped below 1,000. On a monthly scale, sales across

March, April, and May stayed fairly consistent, showing no major seasonal surges. However, when we broke the data down by day of the week, a clear pattern emerged: weekdays, especially Monday, Tuesday, and Friday, tended to perform better, while Sundays saw the least activity.

5.3.2 Justification: These insights are practical for planning and decision-making. Since sales are stronger at the beginning and end of the workweek, it makes sense to focus inventory and staffing efforts around those days. Lower weekend sales, particularly on Sundays, suggest an opportunity to reduce stock or try weekend-specific promotions to boost traffic. The steady monthly demand means stock planning can remain balanced without big seasonal adjustments. Overall, knowing when demand peaks and drops allows for smarter resource use and fewer missed sales opportunities.

5.4 ABC Segmentation Analysis:

5.4.1 Process: Sales data from Amrit Pharmacy was segmented using the ABC analysis method based on each SKU's contribution to total revenue.

5.4.1.1 Category A (High Priority): These are top-selling items making up around 70-80% of revenue, though they are only a small part of total SKUs. For example, Examination Delsweeft Gloves 100 and Stenvita-Gold Tab consistently generate high income. These require monitoring on a regular basis and consistency in stock availability.

5.4.1.2 Category B (Moderate Priority): These are selling items making up around 20-30% of revenue, though they are a small part of total SKUs. In the analysis VOVERAN SR 100MG TAB 10S and CLOVISTA 625 TAB are the category B products. These require these are important products for sale so monitoring on a regular basis is very important and make sure stock availability.

5.4.1.3 Category C (Low Priority): This category items contribute only about 5-10% of total revenue and typically experience slow sales. In amrit pharmacy analysis I found DISPOSABLE NEEDLES G 26 X 0.5 DN2605, GENTICYN 80MG / 2ML INJ and NEOWAY 3 WAY STOPCOCK etc. are the category C items. They often include specialized supplements or medicines with limited demand. Due to their low turnover, these products should be reviewed regularly and maintained in minimal quantities. This approach helps optimize storage space, reduce the risk of expiry-related losses, and improve overall inventory efficiency.

5.4.2 Justification: ABC segmentation allows Arvind Kumar to focus on keeping high-revenue products well-stocked, ensuring customer satisfaction and revenue growth. It helps him manage moderately important items efficiently, with targeted promotions when needed. For slow-moving products, he can minimize inventory to reduce waste and

increase the space in the shop. This approach streamlines inventory management and supports better business decisions.

6. Results and Findings:

6.1 Data Cleaning and Consolidation: When I first opened up the pharmacy's sales data, it honestly looked like someone had just dumped everything in without proper order. There were double entries, some missing info, and a bunch of extra columns that didn't really matter. I went through and cleared out the repeats, filled in any blanks I could, and made sure all the medicine names and dates actually lined up. After that, I gathered up each day's numbers so every product had its own simple daily record, instead of being scattered all over the place. Once things were finally straightened out, it was a lot easier to spot which medicines were moving fast, which ones were just sitting there, and if anything odd was happening with returns. Getting all this sorted was a bit of a hassle, but it made the rest of the work much clearer and less frustrating.

6.2 Descriptive Analysis of Product Movement:

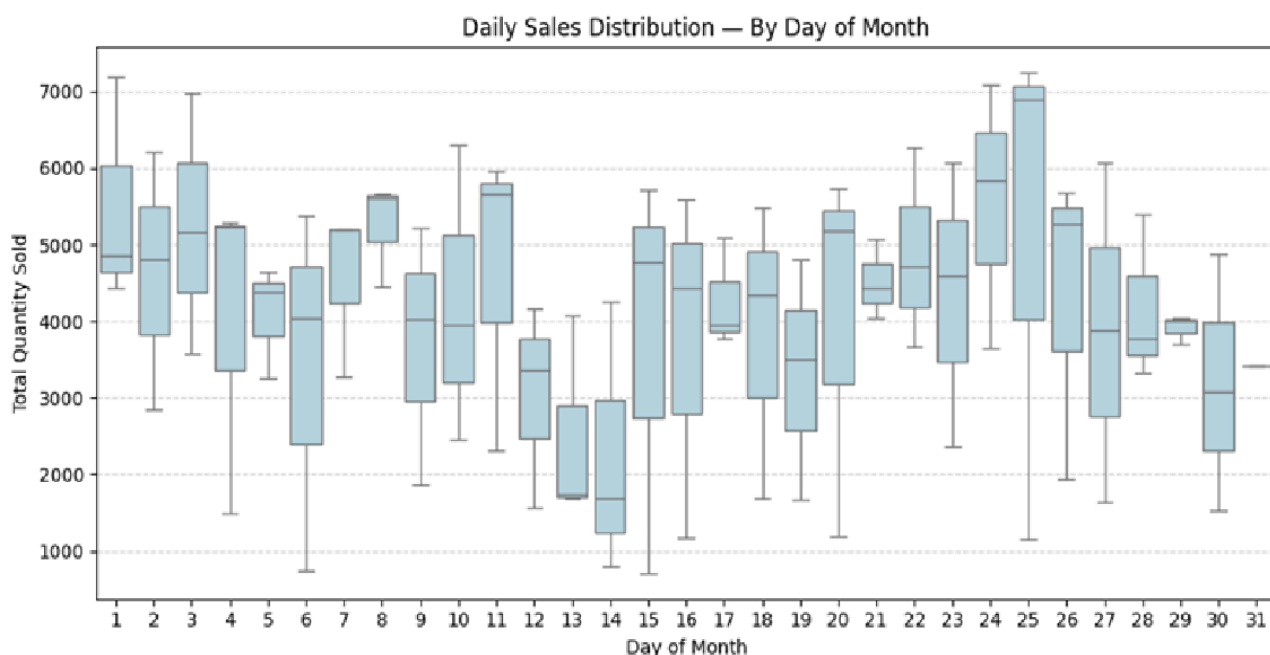


Image 6: Daily Sales Distribution

If you look at this box-plot for March, you can see that sales don't really follow a flat line—they are all over the place. At the very beginning of the month, especially in those first few days, sales are not just high but also jump around a lot from day to day. Then, once you get to the middle of the month, things seem to slow down; the numbers drop and there's less of that big up-and-down swing. But that quiet patch doesn't last. As we head toward the last week, especially between the 23rd and 26th, sales shoot back up and become more

unpredictable again. It kind of looks like the pharmacy is busiest at the start and end of the month, with a bit of a lull in the middle. That is the sort of thing you would want to keep in mind if you're trying to figure out when to order more stock or plan for busier days.

6.3 Category-Level Trend Analysis: The monthly sales trend by category shows distinct movement patterns across departments. Gastroenterology products experienced a steep and continuous rise from March to April, indicating growing demand or bulk institutional consumption. Orthopaedic items remained stable over the three month span, suggesting consistent patient usage. Cardiology sales showed a slight but steady decline, which may point to either inventory adjustments or prescription shifts. Surgical consumables recorded a significant drop, potentially due to returns or reduced procedural usage. Urology items showed mild fluctuation but overall maintained a low-volume footprint. This analysis highlights how departmental behavior influences stock planning and category-level focus areas.

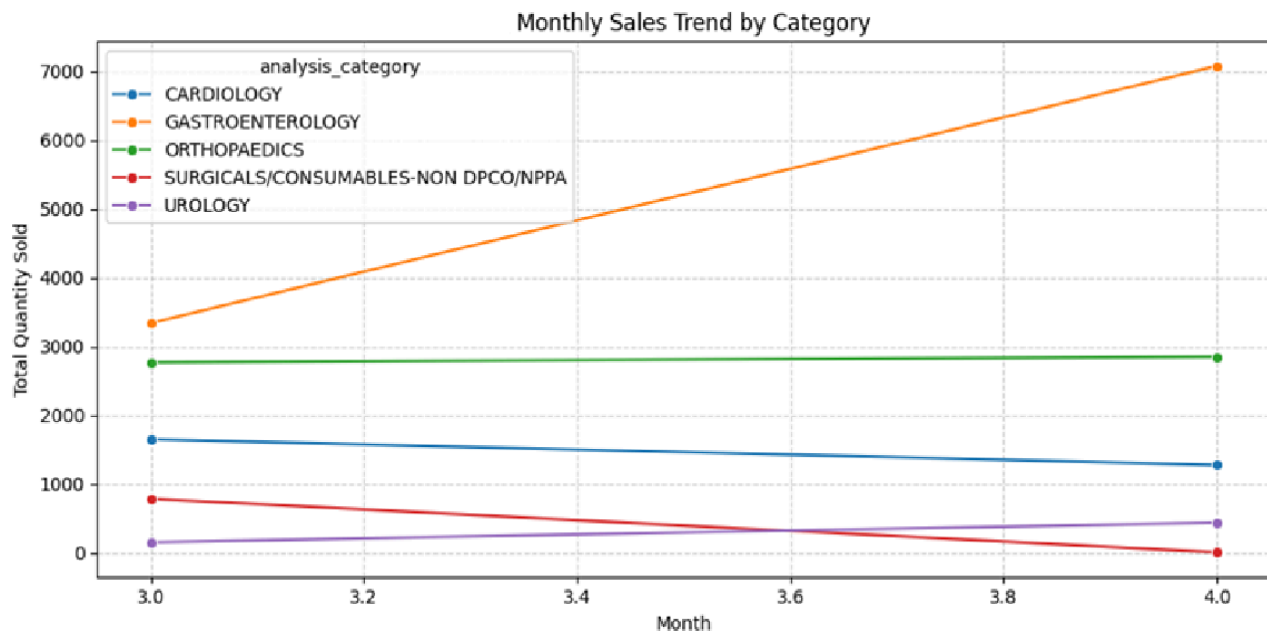


Image 7: Monthly Sales Trend by Category

6.4 Weekday-wise Sales Pattern Analysis: If you look at how sales change over the week, there's a clear pattern. Mondays and Tuesdays are the busiest days, with the pharmacy selling more on these days than any other. It probably has to do with people coming in after the weekend, either with new prescriptions or to restock what they need for the week ahead. The numbers stay pretty strong through Thursday and Friday, with a lot of folks likely picking up medicines before the weekend starts. But once Saturday and especially Sunday roll around, things slow down quite a bit-Sunday is actually the quietest day for sales. So, if you are

planning when to have extra stock or more staff on hand, it makes sense to focus on the start and end of the work week, since that is when most customers show up.

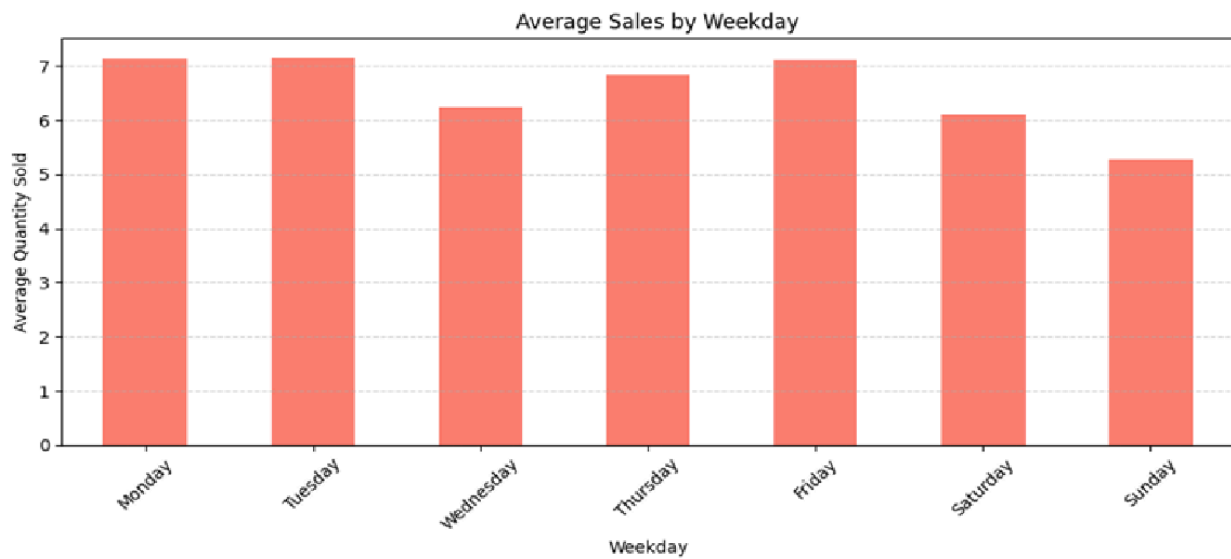


Image 8: Average Sales by Weekdays